

Structure:

- **Early Days:** Initially, Linux file systems were simple and followed the hierarchical directory structure. The root directory (“/”) served as the base, and directories like “/home” stored user files, akin to Windows’ “Documents and Settings.”
- **Extensibility:** Over time, Linux adopted more flexible structures. For instance, the ext2 file system introduced features like inodes, data blocks, and directories. Later versions (ext3, ext4) refined these concepts, allowing for larger file sizes, journaling, and improved performance.

Efficiency:

- **Inodes and Data Blocks:** Inodes track file metadata (permissions, size, location), while data blocks store actual file content. Each file resides in multiple data blocks across the hard drive.
- **Block Sizes:** File systems evolved to optimize block sizes. Larger blocks enhance sequential read/write performance, while smaller blocks reduce wasted space for small files.
- **Journaling:** Journalized file systems (e.g., ext3, ext4) improved crash recovery by maintaining a transaction log. This balances efficiency with data integrity.

Security:

- **Access Control:** Linux file systems enforce permissions (read, write, execute) at the file and directory level. Users, groups, and others have distinct access rights.
- **Extended Attributes:** Modern file systems support extended attributes (xattrs) for additional metadata (e.g., SELinux labels). These enhance security and context.
- **Encryption:** File system encryption (e.g., dm-crypt, LUKS) safeguards data at rest. It prevents unauthorized access even if physical storage is compromised.